

Name:
Period:

Earthquakes & the Mercalli Scale

An earthquake is a sudden, violent movement in the Earth's crust which may cause extensive damage and loss of life. The amount of damage caused by an earthquake depends on the intensity of the quake and the distance from the epicenter. Other factors that are important are building materials, type of construction, and the condition of the ground. Buildings with foundations in solid rock usually suffer less damage than those built on Sand or landfills.

In this activity, you will use data from an earthquake that occurred in New York State. You will determine the Mercalli rating based on the damage reported and construct an isoseismal map in order to find the epicenter and evaluate the damage to the surrounding region.

Problem:

1. How can the amount of damage caused by an earthquake be described?
2. How can the Mercalli scale be used to determine the location of an earthquake?

Procedure:

1. Using the 1010 WINS Radio Report, determine the Mercalli reading each location in the New York state area by comparing the caller's description to the Mercalli scale.
2. Record the number of the Mercalli rating next to each location on the New York state map.
3. Construct an isoseismal map by drawing lines that connect equal Mercalli values.
4. Answer the *Discussion Questions*, on the next pages

Discussion Questions:

1. Based on your map, where is the most damage? Explain how you arrived at this conclusion.

Answer:

2. Using ESRT page 3, what is the approximate latitude and longitude of the epicenter? Round your answer to the nearest tenth of a degree.

Answer:

3. Why might some buildings near the epicenter sustain less damage than others?

Answer:

4. Explain what factors caused the Mercalli intensity values from any earthquake to vary from place to place.

Answer:

5. Draw and label a graph showing the relationship between distance and damage reported.

Answer:

6. Do you think the Mercalli scale is adequate for distinguishing an extremely strong earthquake that occurred far away from a less severe event that occurred close by? Explain your answer.

Answer:

7. What is the name of the other scale used to identify an earthquake's magnitude?

Answer:

8. Where is the earthquake hazard higher, in New York or California? Explain.

Answer:

9. Research the latest earthquakes to affect New York State. List the last three earthquakes.

	Earthquake 1	Earthquake 2	Earthquake 3
Location			
Date			
Magnitude (#)			